Christos Hadjinikolis

Engineering Manager | Production ML & Real-Time Data Systems
Ph.D. Computer Science | Engineering Leadership, Streaming Systems & AI Standards
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profiles
website
LinkedIn
GitHub
Google Scholar

leadership
6 direct reports
10-person team
hiring/coaching
delivery ownership

systems
real-time data
Kafka/Flink
model/API serving
observability

ML product
production ML
eval/replay
data contracts
model monitoring

maturity
ADRs/forums
runbooks/on-call
local E2E tests
rollback/fallback

domain
commodity intel
maritime data
AI standards
explainability

tooling
PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

I am an **Engineering Manager and ML systems lead** with 16 years across applied AI, data platforms, backend services, consulting, and academic research. I lead cross-functional teams that turn research-grade models, high-volume data, and ambiguous product requirements into reliable production systems. My recent work combines people management, client and stakeholder communication, architecture, delivery ownership, engineering practice, streaming infrastructure, model/API serving, evaluation/replay loops, observability, and reliability for client-facing intelligence products.

work-experience

Vortexa Ltd, London, UK 12/2020–present — Engineering Manager | ML Systems Lead

I lead technical direction and delivery for Vortexa's live post-ingestion ML/data estate, turning **6M** vessel-position records/hour into production commodity intelligence for **13.5K** monitored vessels. Manage **6** direct reports and lead a **10-person** cross-functional team accountable for model quality, data-product interfaces, client and stakeholder clarity, reliability, and delivery maturity.

- **People & Delivery:** Manage **6** direct reports across Product, SME analysis, Data Science, and Data Engineering; own performance/career development, hiring for **6** roles, delivery accountability, and contributed to Data Production growth from **4 to 30+**.
- **Technical Direction:** Shape architecture and engineering direction across model/API serving, streaming pipelines, evaluation/replay interfaces, data contracts, and downstream product integrations; serve as a senior technical escalation point while remaining close to implementation and long-term system evolution.
- **Production ML Systems:** Led research-to-production delivery for destination and arrival-time sequence/transformer models in **PyTorch**; established MLflow, model/data versioning, automated evaluation gates, replay, monitoring, and model-serving workflows at **6M** events/hour.
- **Streaming & Distributed Processing:** Led the **Kafka Streams-to-Flink** migration as a strategic platform move, improving operational visibility, state/checkpointing, maintainability, and compute scaling independent of Kafka partitioning.
- **Reliability & Observability:** Maintained signal-quality controls, live accuracy monitoring, runbooks, rollback/fallback paths, shared on-call, and **MTTR** under **30** min for production prediction and streaming systems.
- **Engineering Maturity:** Standardised I/O seams, schema/data validations, repo archetypes, ADRs, dev containers, local E2E tests, and DORA/Jira delivery-health signals; cut a **3h** DAG feedback loop to under **5** min.
- **Client & Stakeholder Alignment:** Addressed direct client and stakeholder concerns and ran workshops across Product, SMEs, analysts, and engineers to balance product pull with engineering trade-offs, turning model-quality disputes into measurable workstreams.

DataReply, London, UK 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant, operating inside and contributing to a practice that grew from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped opportunities, agreed deliverables/timelines, managed expectations, placed consultants, and helped build the consulting culture around clear technical delivery across UBS, Vodafone, CNHi, and other clients.

- **Vodafone Group:** Led multi-workstream GCP data-science platform work based on Kubeflow; drove requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance (Azure/Databricks); translated stakeholder needs into DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.

KCL, UCL, David Game College, UK 2010–2016 — Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

education

2010–2014 **Ph.D. in Computer Science** King's College London, UK

***Thesis**: Persuasion Dialogues & Opponent Modelling.*

Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Received **Best Poster Award** at IJCAI 2013 (413 submissions), **EPSRC PhD Scholarship**, **Outstanding TA Award**, and **Graduate Certificate in Academic Practice**.

Access Publications on Google Scholar

2004–2010 **Diploma (BEng) in Computer Engineering** University of Thessaly, GR

Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence***.

2002–2004 **Military Officer: Second Lieutenant** National Guard, CY

Completed national service as a commissioned Second Lieutenant.

selected AI/platform tooling

Promet Local AI harness

Applied AI project exploring voice, memory, tool-use, and streaming assistant workflows; hands-on work on local runtimes, faster-whisper STT, Kokoro/**Hugging Face** assets, tool routing, schema validation, approval gates, durable state, traces, and deterministic replay/evaluation for auditable LLM interactions.

PyPI **skeleton-replay** Runtime evidence

Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and evidence-led code understanding.

JetBrains **Skeleton Replay plugin** IDE workflow

Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.

PyPI **dynamicio** Data contracts

Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.

committees/affiliations

since 10/2024 **University College London, Dept. of Information Studies** Associate Researcher

Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.

since 01/2021 **ISO - International Organisation for Standardisation** Committee Expert Member for JTC'21

Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.

talks/interviews

2023 **Bill Raymonds, Agile in Action** Podcast Interview

Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, **Details:** YouTube

2022 **Open Data Science Conference** Industry Conference Talk

dynamicio, a published **PyPI library** for abstracting I/O in ML/data workflows, **Details:** here

2020 **iunera: Big Data Innovators** Interview Blog Post

The Agile approach in data science explained by an ML expert, **Details:** here

2020 **Big Data Warsaw** Industry Conference Talk

Monitoring & Analysing Communication and Trade Events as Graphs, **Details:** here

2018 **Connected Data London** Panelist @ Industry Conference

AI tribes and interconnections: what's graph got to do with it?, **Details:** here

2018 **Minds Mastering Machines** Industry Conference Talk

A talk on doing Data Science the Agile Way, **Details:** here

certificates & awards

2017–2020 **Selected ML/cloud credentials** AWS, Google Cloud, O'Reilly

AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.

2016–2019 **Data/graph engineering credentials** Coursera, Elastic, Neo4j

Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.